

CSE 4631

# Digital Signal Processing

Week 7

# Importance of Fourier Decomposition

- The goal of decomposition is to end up with something *easier* to deal with than the original signal
  - For example, impulse decomposition allows signals to be examined one point at a time, leading to the powerful technique called convolution.
- The component sine and cosine waves are simpler than the original signal because they have a property that the original signal does not have: *sinusoidal fidelity*
- **Sinusoidal fidelity:** A sinusoidal input to a linear system is guaranteed to produce a sinusoidal output. Only the amplitude and phase of the signal can change; the frequency and wave shape must remain the same.

# Family of Fourier Transform

Can be broken into four categories, resulting from the four basic types of signals that can be encountered:

- Aperiodic-Continuous  $\longrightarrow$  Fourier Transform
- Periodic-Continuous  $\longrightarrow$  Fourier Series
- Aperiodic-Discrete  $\longrightarrow$  Discrete-Time Fourier Transform
- Periodic-Discrete  $\longrightarrow$  Discrete Fourier Transform (DFT)

All four classes of signals extend to positive and negative *infinity*. And there isn't any version of the Fourier Transform that uses finite length signals.

# Family of Fourier Transform (contd.)

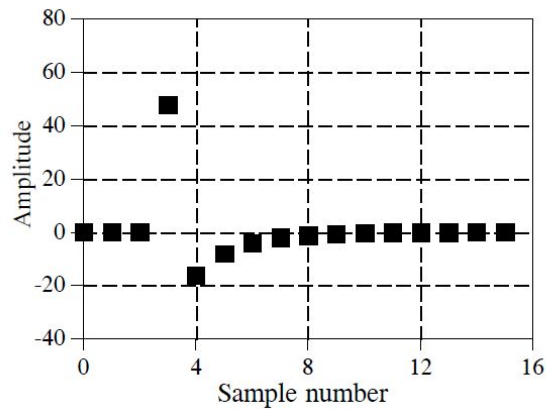
So, we need to make the finite data look like an infinite length signal. This is done by “imagining” that the signal has an infinite number of samples on the left and right of the actual points.

- All the imagined samples have a value of zero  $\longrightarrow$  the signal looks *discrete* and *aperiodic*  $\longrightarrow$  Discrete Time Fourier Transform
- All the imagined samples are duplications of actual samples  $\longrightarrow$  the signal looks *discrete* and *periodic*  $\longrightarrow$  Discrete Fourier Transform (DFT)

It turns out, an infinite number of sinusoids are required to synthesize a signal that is *aperiodic*. Hence, the **only type of Fourier Transform that can be used in DSP is DFT**. In other words, digital computers can only work information which is ***discrete*** and ***finite*** in length.

# Notation and Format of the Real DFT

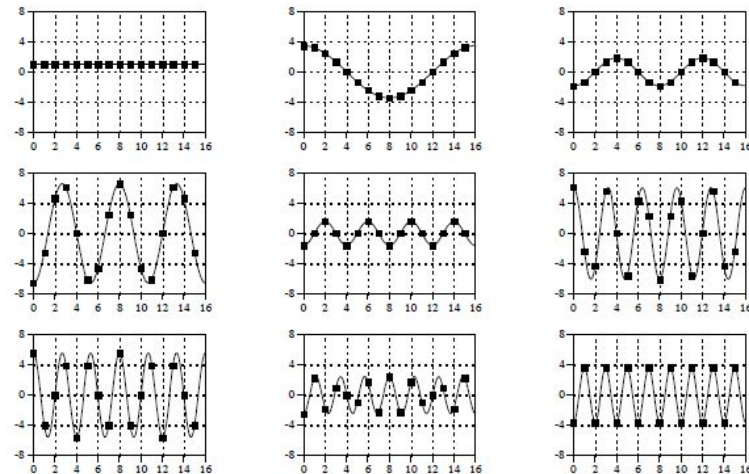
- DFT changes an  $N$  point input signal into two  $N/2+1$  point output signals
- The output signals contain the amplitudes of the component sine and cosine waves
- The input signal is said to be in **time domain**
- The term **frequency domain** is used to describe the amplitudes of the sine and cosine waves
- Given the time domain signal, the process of calculating the frequency domain is called **decomposition, analysis**, the **forward DFT**, or simply **the DFT**.
- The opposite of this is called **synthesis**, or the **inverse DFT**



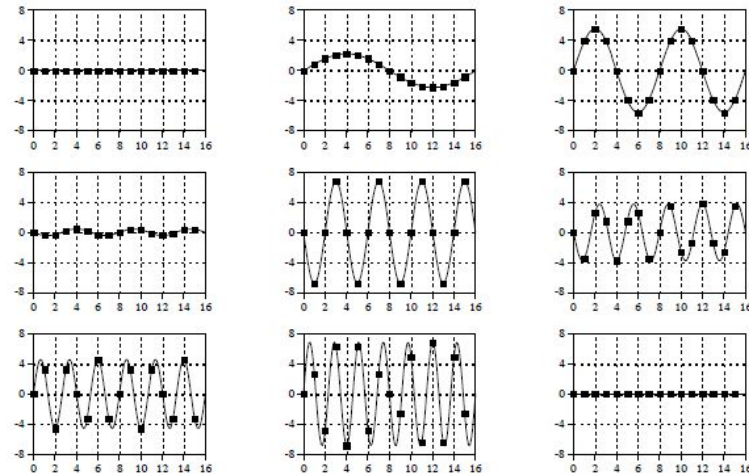
DECOMPOSE

SYNTHESIZE

Cosine Waves



Sine Waves



# Notation and Format of the Real DFT (contd.)

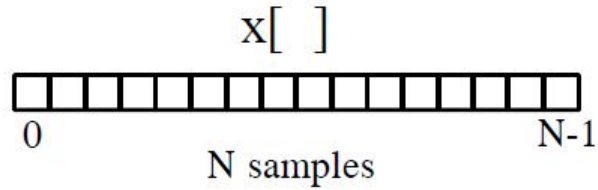
- The number of samples in the time domain is usually represented by the **variable  $N$**
- A power of two is usually chosen for  $N$ . Two reasons:
  - Digital data storage uses binary addressing
  - The most efficient algorithm for calculating the DFT, the Fast Fourier Transform (FFT), usually operates with  $N$  that is a power of two.
- Standard DSP notation uses lower case letters to represent time domain signals, such as  $x[ ]$ ,  $y[ ]$ , and  $z[ ]$
- The corresponding upper case letters used to represent their frequency domain, that is,  $X[ ]$ ,  $Y[ ]$ , and  $Z[ ]$

# Notation and Format of the Real DFT (contd.)

Assume that the frequency domain of an  $N$  point time domain signal  $x[n]$  is  $X[k]$ .

- Then  $X[k]$  is an array of  $N/2+1$  samples. These are called Real part of  $X[k]$ :  $Re\{X[k]\}$  and the Imaginary part of  $X[k]$ :  $Im\{X[k]\}$
- The values in  $Re\{X[k]\}$  are the amplitudes of the cosine waves, while the values in  $Im\{X[k]\}$  are the amplitudes of the sine waves.

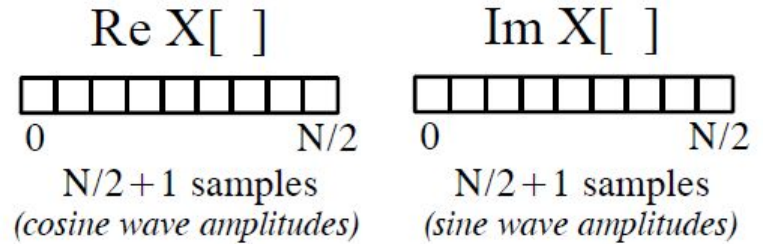
## Time Domain



Forward DFT

Inverse DFT

## Frequency Domain



collectively referred to as  $X[ \ ]$

# The Frequency Domain Independent Variables

The horizontal axis of the frequency domain can be referred to in **four different ways**:

- Labeled from 0 to  $N/2$  samples in the arrays
  - Indexes are integer
  - Index used with this labeling is  $k$ :  $Re X[k]$  and  $Im X[k]$ , where  $k$  runs from 0 to  $N/2$  in steps on one
- Labeled as fraction of the *sampling rate*. Meaning that the values always run between 0 and 0.5
  - Index used with this labeling is  $f$  (for frequency) :  $Re X[f]$  and  $Im X[f]$ , where  $f$  takes on  $N/2+1$  equally spaced values between 0 and 0.5
  - Conversion from first notation:  $f = k/N$
  - Advantage:
    - It's unitless, so the DFT result is general-purpose and portable.
    - Not tied to a specific sampling rate
    - Shows the frequency components without needing to know the actual sampling rate.

# The Frequency Domain Independent Variables (contd.)

- Third style is similar to the second, except the horizontal axis is multiplied by  $2\pi$ 
  - Index used with this labeling is  $\omega$ :  $Re X[\omega]$  and  $Im X[\omega]$ , where  $\omega$  takes on  $N/2+1$  equally spaced values between 0 and  $\pi$
  - $\omega$  is called the natural frequency
  - $\omega$  has the units of radians.
- Fourth method is to label in terms of the analog frequencies used in a particular application
  - For example, if the system is being examined has a sampling rate of 10 kHz, graphs of the frequency domain would run from 0 to 5 kHz.
  - Advantage: Presents the frequency data in terms of a *real world* meaning
  - Disadvantage: Tied to a specific sampling rate

A cosine function would be written using the first three notations like this:

1. Using  $k$ :  $\cos(2\pi kn/N)$
2. Using  $f$ :  $\cos(2\pi fn)$
3. Using  $\omega$ :  $\cos(\omega n)$

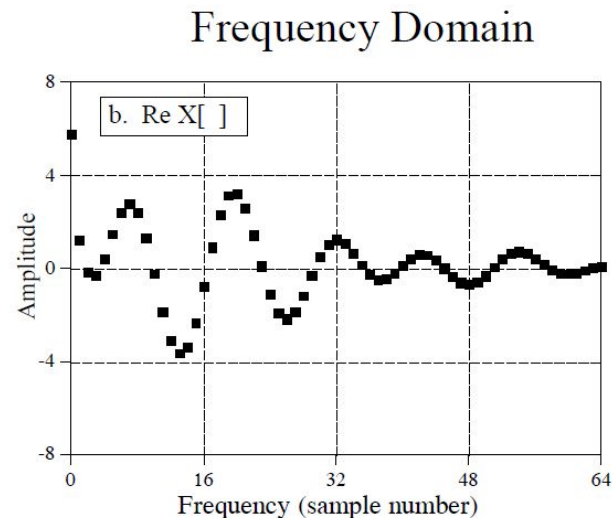
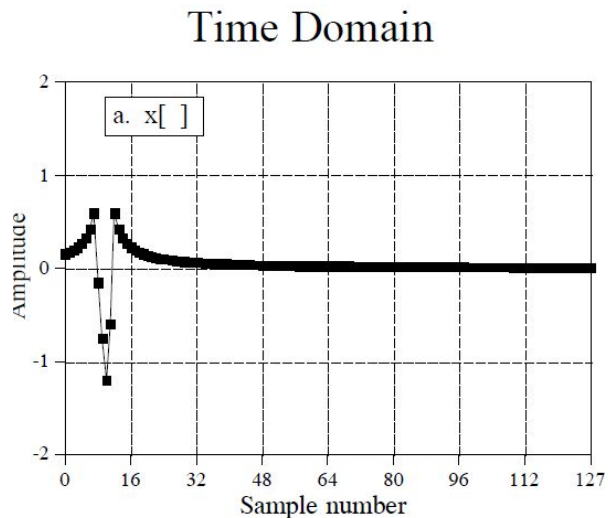
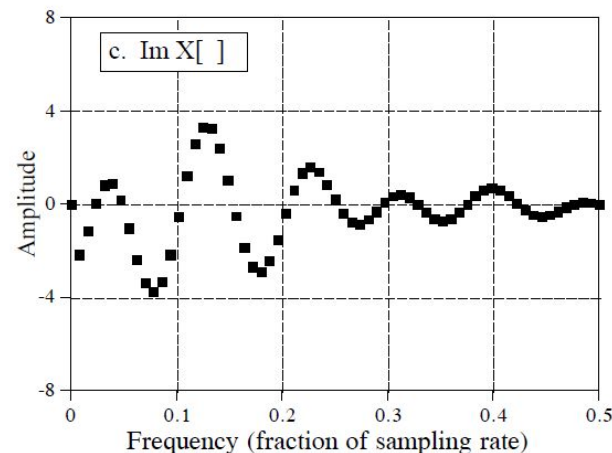


FIGURE 8-4

Example of the DFT. The DFT converts the time domain signal,  $x[n]$ , into the frequency domain signals,  $\text{Re } X[k]$  and  $\text{Im } X[k]$ . The horizontal axis of the frequency domain can be labeled in one of three ways: (1) as an array index that runs between 0 and  $N/2$ , (2) as a fraction of the sampling frequency, running between 0 and 0.5, (3) as a natural frequency, running between 0 and  $\pi$ . In the example shown here, (b) uses the first method, while (c) use the second method.



# DFT Basis Functions

- The sine and cosine waves used in the DFT are commonly called the DFT basis functions
- The basis functions are a set of sine and cosine waves with *unity* amplitude.
- If you assign each amplitude (the frequency domain) to the proper sine or cosine wave (the basis functions), the result is a set of *scaled* sine and cosine waves that can be added to form the time domain signal
- The DFT basis functions are generated from the equations:

$$c_k[i] = \cos(2\pi ki/N)$$

$$s_k[i] = \sin(2\pi ki/N)$$

$c_k[i]$  and  $s_k[i]$  are the cosine and sine waves, each  $N$  points in length, running from  $i = 0$  to  $N-1$ . The parameter,  $k$ , determines the frequency of the wave, taking on values between 0 and  $N/2$ .

$c_k[\ ]$  is the cosine wave for the amplitude held in  $Re\ X[\ ]$ , and  $s_k[\ ]$  is the sine wave for the amplitude held in  $Im\ X[\ ]$

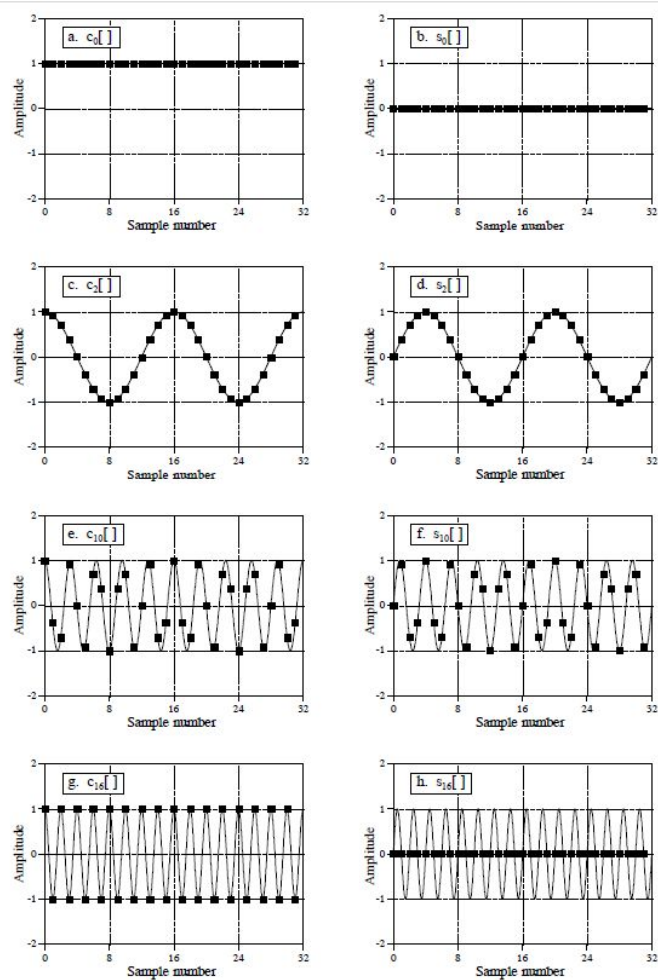


FIGURE 8-5

DFT basis functions. A 32 point DFT has 17 discrete cosine waves and 17 discrete sine waves for its basis functions. Eight of these are shown in this figure. These are discrete signals; the continuous lines are shown in these graphs only to help the reader's eye follow the waveforms.